6-6. Multiple regression: Example 6.5 Scottish hill races data

These data are made available in R using

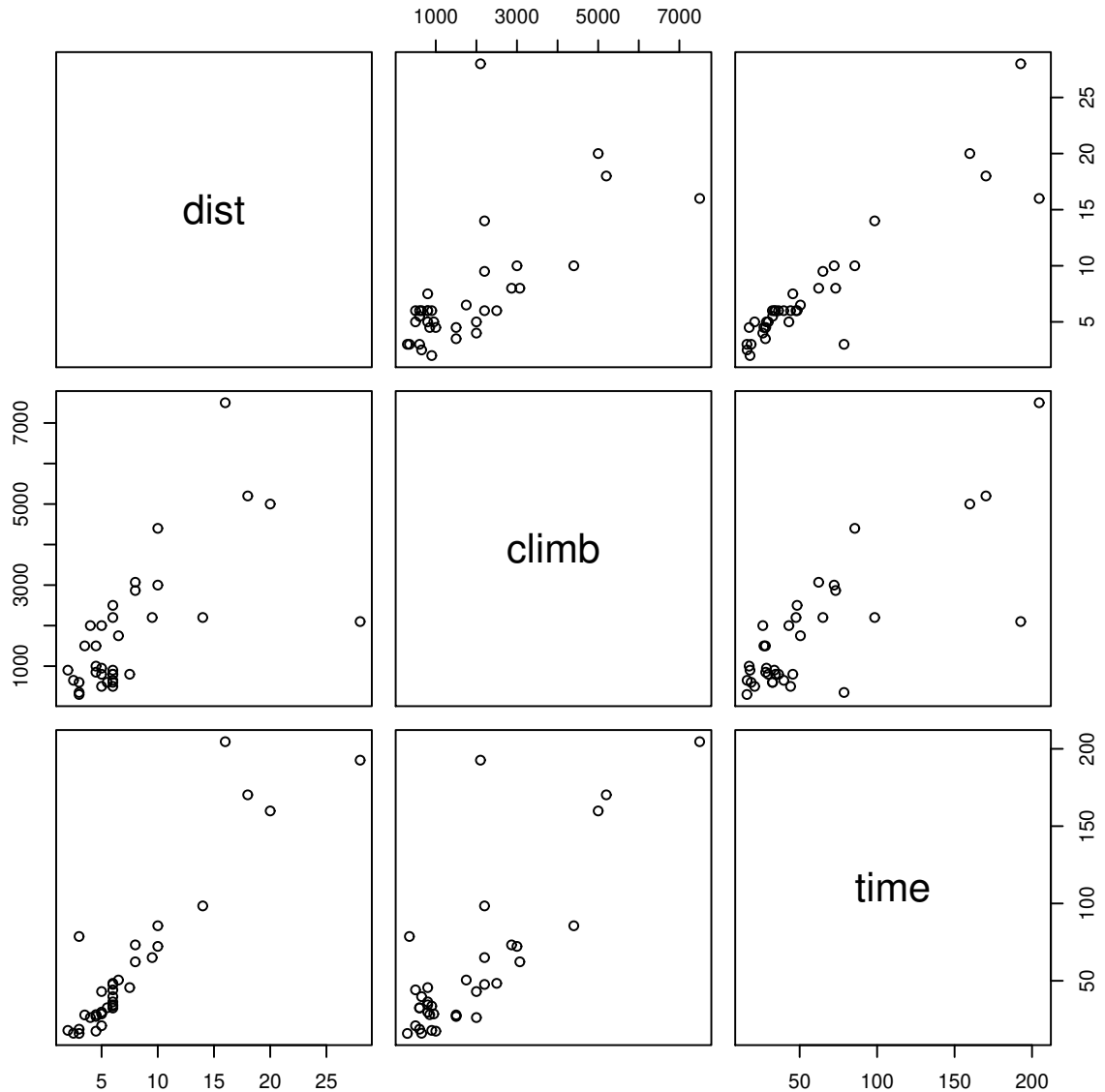
```
> library(MASS); data(hills); attach(hills)
```

They record *time* (minutes) in 1984 of 35 Scottish hill races, against *distance* (miles) and total *height climbed* (feet).

We regard *time* as the response variable, and seek to model how its conditional distribution depends on the explanatory variables *distance* and *climb*.

```
> hills
                dist climb    time
Greenmantle      2.5   650  16.083
Carnethy         6.0  2500  48.350
Craig Dunain     6.0   900  33.650
Ben Rha          7.5   800  45.600
Ben Lomond       8.0  3070  62.267
Goatfell         8.0  2866  73.217
Bens of Jura    16.0  7500 204.617
Cairnpapple      6.0   800  36.367
Scolty           5.0   800  29.750
Traprain         6.0   650  39.750
Lairig Ghru     28.0  2100 192.667
Dollar           5.0  2000  43.050
Lomonds          9.5  2200  65.000
Cairn Table      6.0   500  44.133
Eildon Two       4.5  1500  26.933
Cairngorm       10.0  3000  72.250
Seven Hills     14.0  2200  98.417
Knock Hill       3.0   350  78.650
Black Hill       4.5  1000  17.417
Creag Beag       5.5   600  32.567
Kildcon Hill     3.0   300  15.950
Meall Ant-Suidhe 3.5  1500  27.900
Half Ben Nevis   6.0  2200  47.633
Cow Hill         2.0   900  17.933
N Berwick Law    3.0   600  18.683
Creag Dubh       4.0  2000  26.217
Burnswark        6.0   800  34.433
Largo Law        5.0   950  28.567
Criffel          6.5  1750  50.500
Acmony           5.0   500  20.950
Ben Nevis       10.0  4400  85.583
Knockfarrel      6.0   600  32.383
Two Breweries   18.0  5200 170.250
Cockleroi        4.5   850  28.100
Moffat Chase    20.0  5000 159.833
```

```
> pairs(hills)
```



Plots show that the response variable *time* has a strong positive association with each of the explanatory variables *distance* and *climb* – although a stronger dependence on *distance*.

However, the two explanatory variables *distance* and *climb* also have a strong positive association with each other, and this **complicates the modelling**.

Preliminary analysis of the data suggests that the observation number 18, corresponding to Knock Hill, is likely to be in error (use `plot` and `identify` commands) – the time is much too great for the given distance and climb, and it may have been misrecorded by 1 hour.

We therefore omit Knock Hill from the analysis. [In practice, be very careful of omitting data points without a strong reason; we do it here to illustrate the method.]

On physical grounds we attempt to find a model with zero intercept.

We consider first a linear model (Model 1) involving both explanatory variables:

$$\text{time} \quad = \quad a \times \text{distance} + b \times \text{climb} + \epsilon$$

```
> hills.model.1 = lm(time ~ -1 + dist + climb, data = hills[-18, ])
> summary(hills.model.1)
Call:
lm(formula = time ~ -1 + dist + climb, data = hills[-18, ])

Residuals:
    Min      1Q  Median      3Q     Max
-17.824  -9.977  -5.554  -2.939  37.360

Coefficients:
      Estimate Std. Error t value Pr(>|t|)
dist  5.467760   0.421492  12.972 2.74e-14 ***
climb 0.010636   0.001618   6.573 2.09e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 11.76 on 32 degrees of freedom
Multiple R-squared:  0.9774,    Adjusted R-squared:  0.976
F-statistic: 693.5 on 2 and 32 DF,  p-value: < 2.2e-16
```

The fitted model is

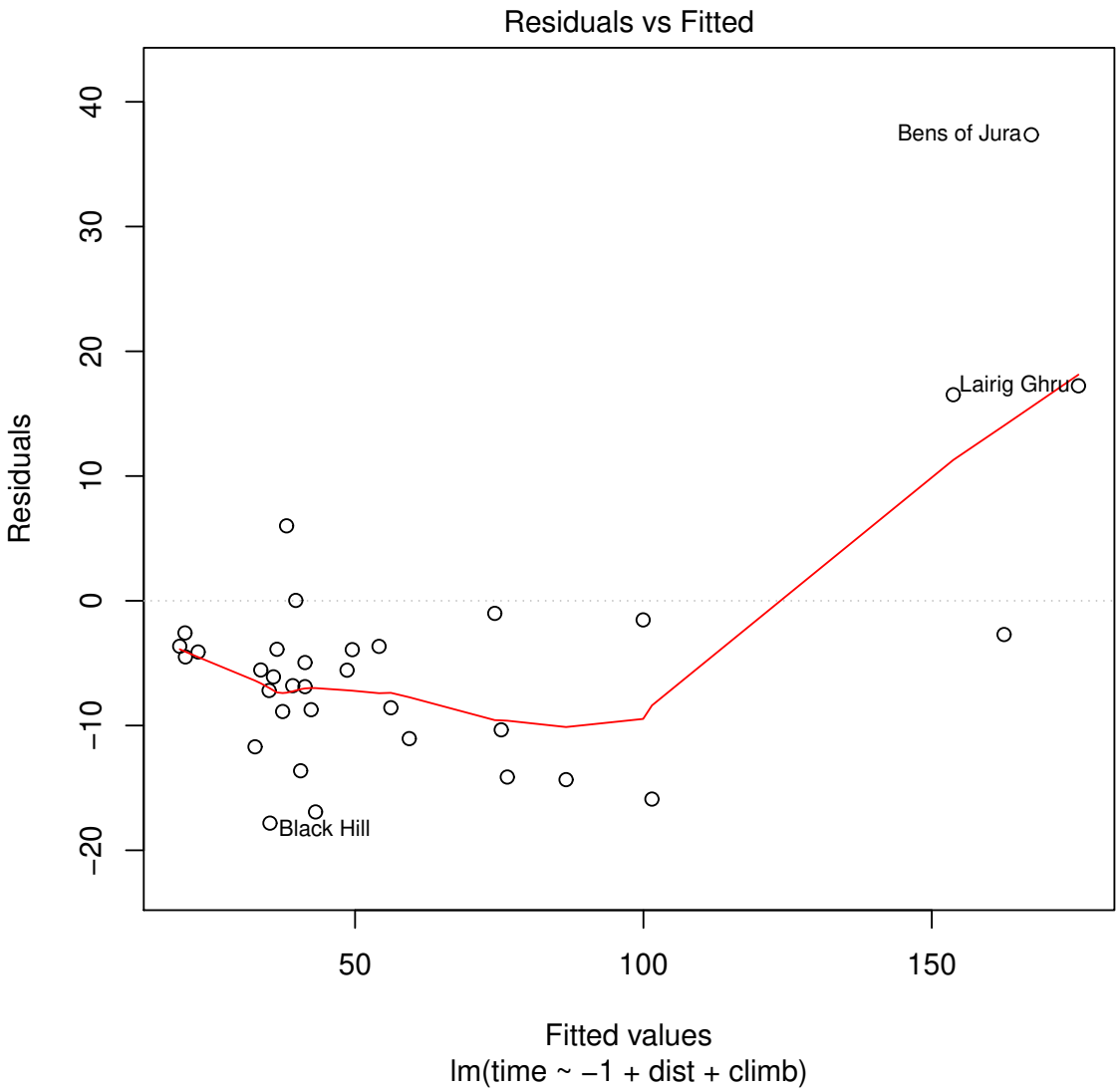
$$\widehat{\text{time}} \quad = \quad 5.47 \times \text{dist} + 0.0106 \times \text{climb}$$

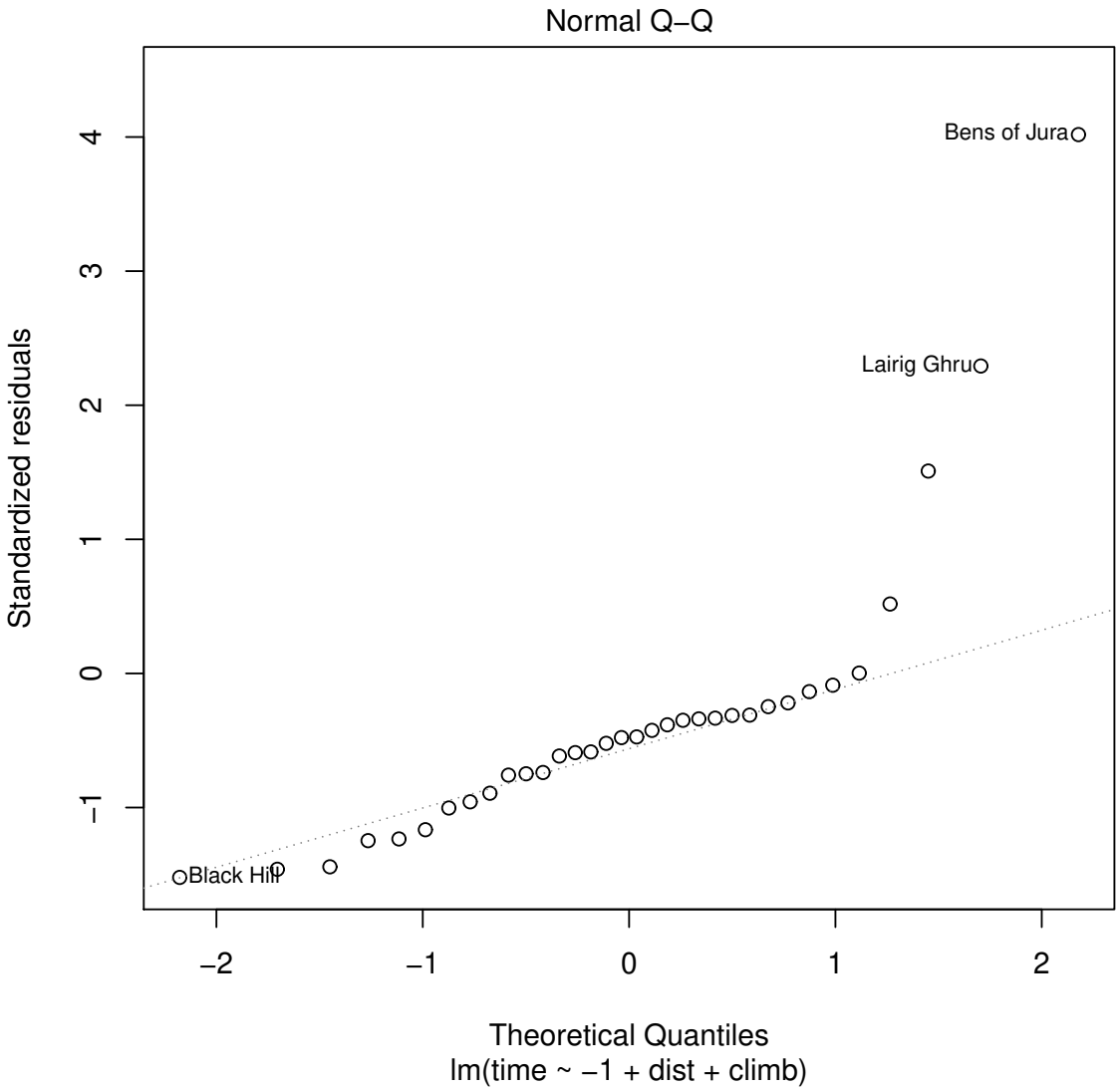
The analysis, in particular the p-values associated with the estimates of the coefficients, shows that distance and climb are both important explanatory variables.

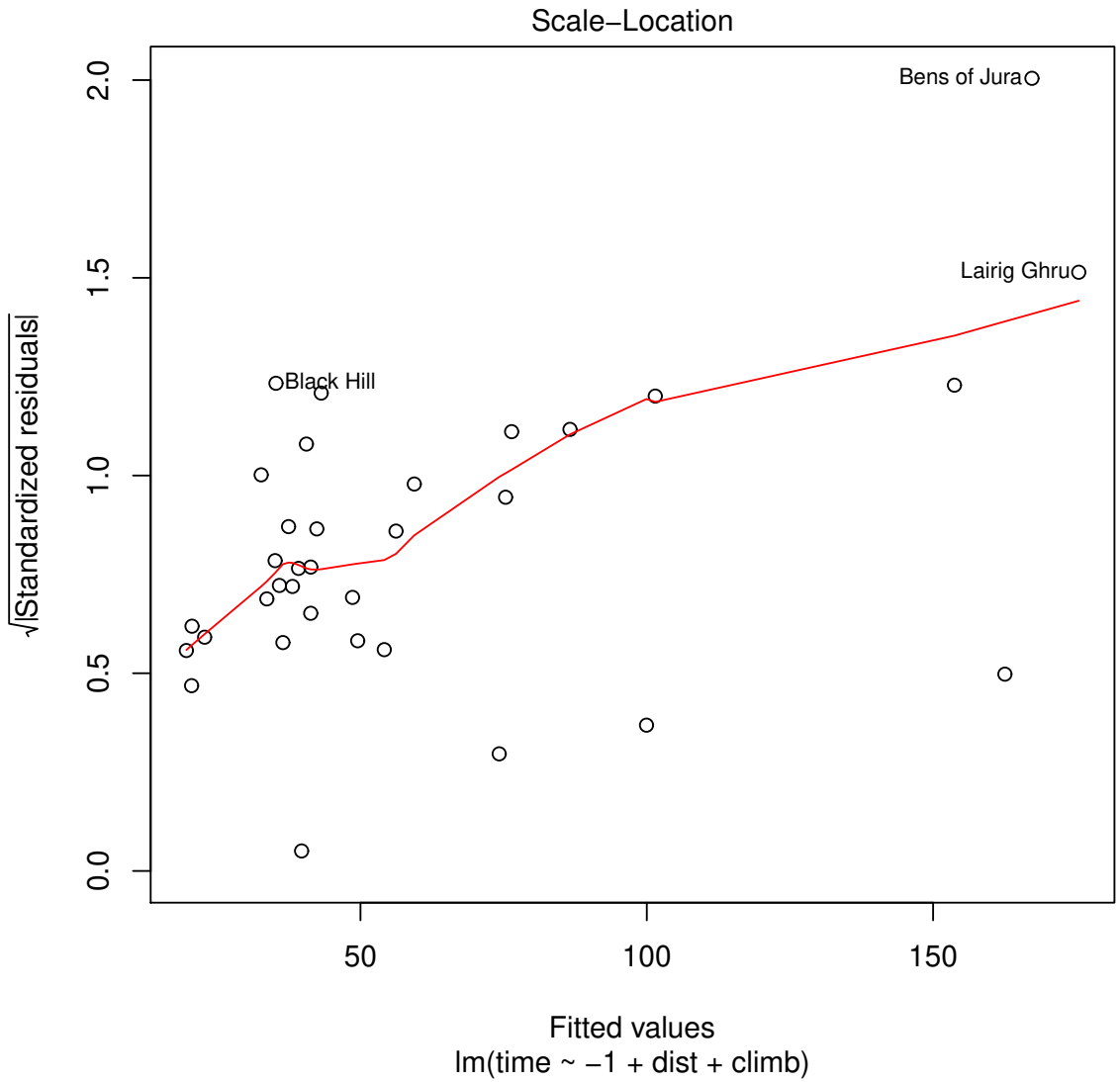
(This can be confirmed by noting the very much poorer fits obtained if either of these variables is omitted.)

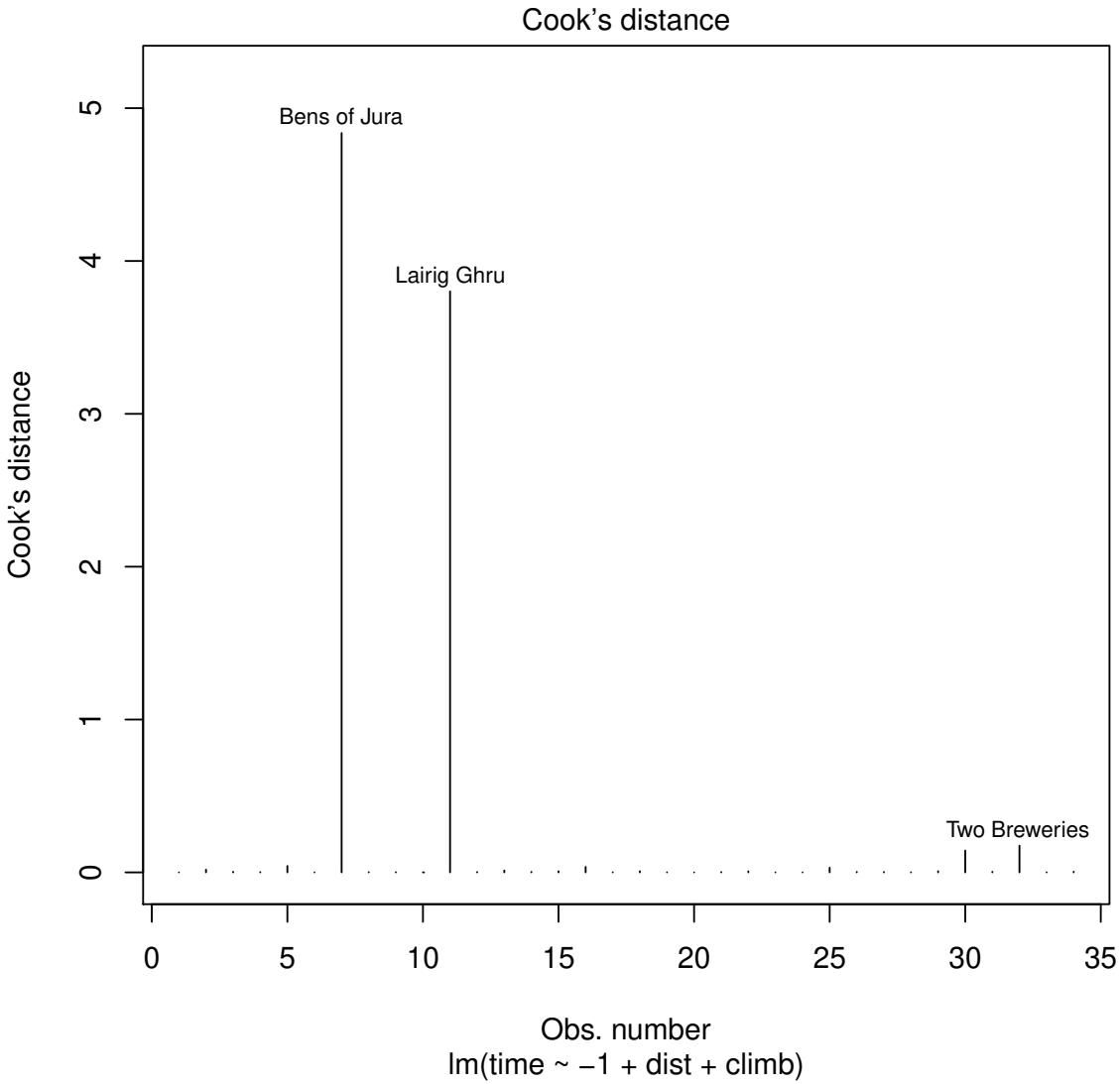
Note: When a model without intercept is fitted, `R` uses a different formula to compute R^2. See `?summary.lm` for details. The R^2 value cannot be used to compare a model with intercept and a model without intercept.

```
> plot(hills.model.1,which=1:4)
```




Normal Q–Q
Standardized residuals
Theoretical Quantiles
lm(time ~ –1 + dist + climb)
Bens of Jura
Lairig Ghru
Black Hill
4
3
2
1
0
–1
–2
–1
0
1
2


Scale−Location
Bens of Jura
Lairig Ghru
Black Hill
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
50
100
150
Fitted values
lm(time ~ −1 + dist + climb)


Cook's distance
Bens of Jura
Lairig Ghru
Two Breweries
Cook's distance
0
1
2
3
4
5
0
5
10
15
20
25
30
35
Obs. number
lm(time ~ −1 + dist + climb)

The Cook's distance plot shows that the observations for Bens of Jura and Lairig Ghru both have a very large influence on the fit.

These observations have the largest values of climb and distance respectively.

We are led to suspect that there may be some nonlinear dependence on climb and/or distance. This would be physically quite natural.

It seems reasonable to introduce quadratic terms as a first attempt to model any nonlinearity.

Consider now the (quite elaborate) Model 2:

$$\begin{aligned} \text{time} &= a_0 \times \text{distance} + b_0 \times (\text{distance})^2 \\ &\quad + c_0 \times \text{climb} + d_0 \times (\text{climb})^2 + \epsilon \end{aligned}$$

```
> hills.model.2 = lm(time ~ -1 + dist + I(dist^2) + climb + I(climb^2),
      data = hills[-18,])
> summary(hills.model.2)
```

Use aov to determine the significance of the covariate, not summary t val

```
Call:
lm(formula = time ~ -1 + dist + I(dist^2) + climb + I(climb^2),
    data = hills[-18, ])

Residuals:
     Min       1Q   Median       3Q      Max
-11.8950  -2.8385   0.1951   3.4281  10.6220

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
dist       5.620e+00  5.477e-01  10.260 2.51e-11 ***
I(dist^2)  3.230e-02  1.797e-02   1.797   0.0823 .
climb      2.621e-04  2.171e-03   0.121   0.9047
I(climb^2) 1.804e-06  2.645e-07   6.820 1.45e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.632 on 30 degrees of freedom
Multiple R-squared:  0.9951,    Adjusted R-squared:  0.9945
F-statistic:  1539 on 4 and 30 DF,  p-value: < 2.2e-16
```

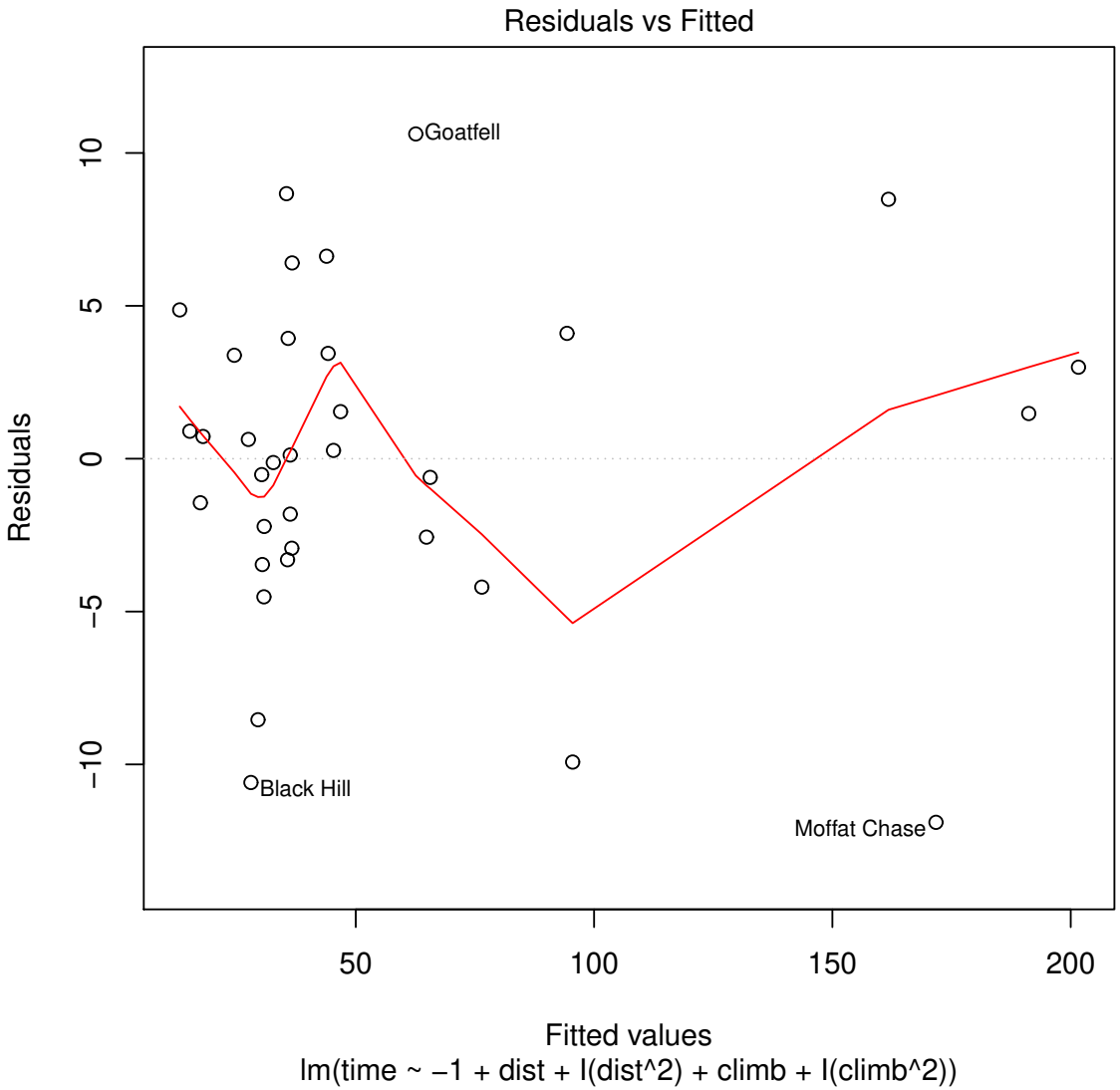
When doing aov, the order of the covariates are important. Should always arrange it according to Sum Squares from high to low.

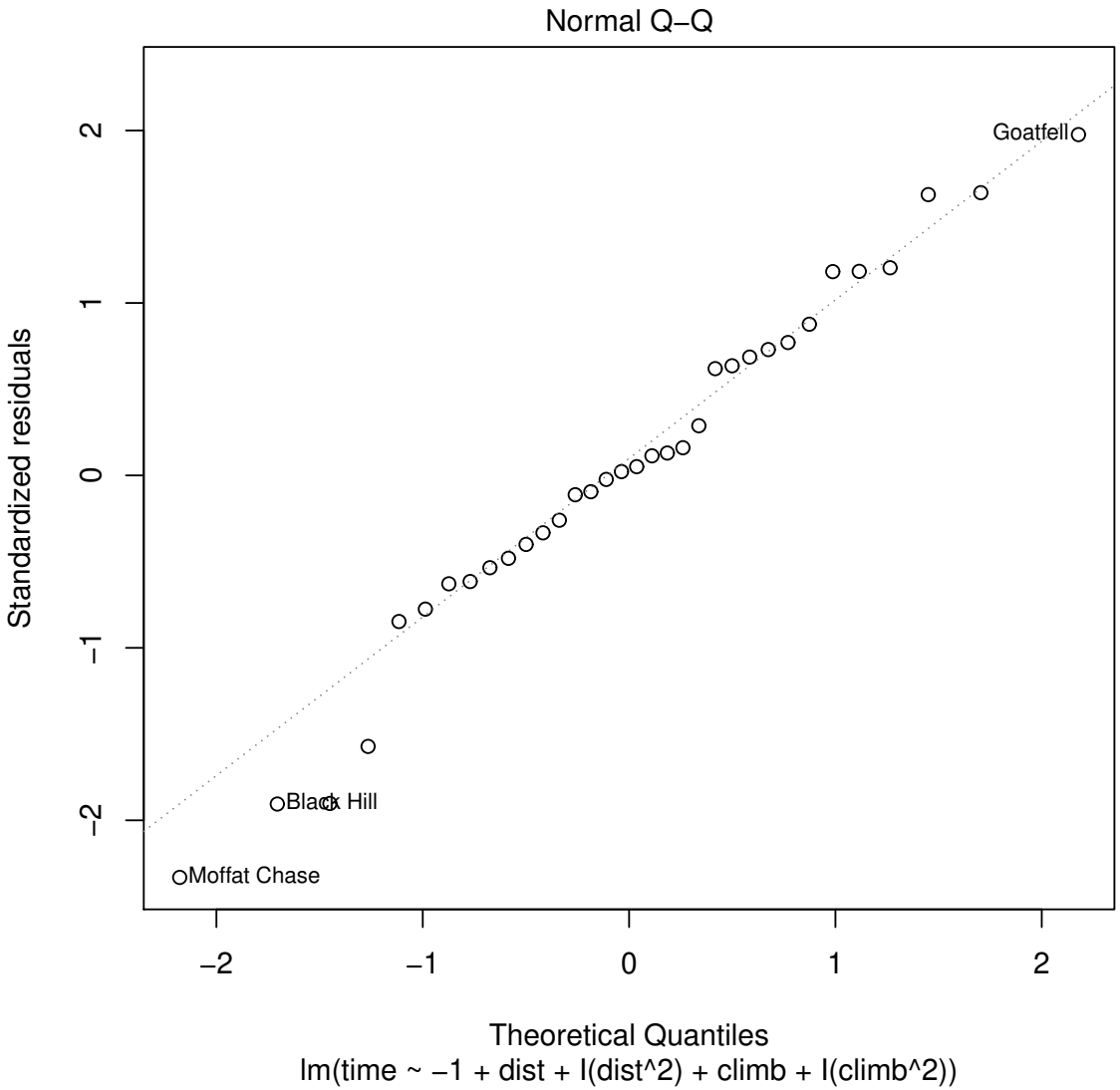
The fitted model is now

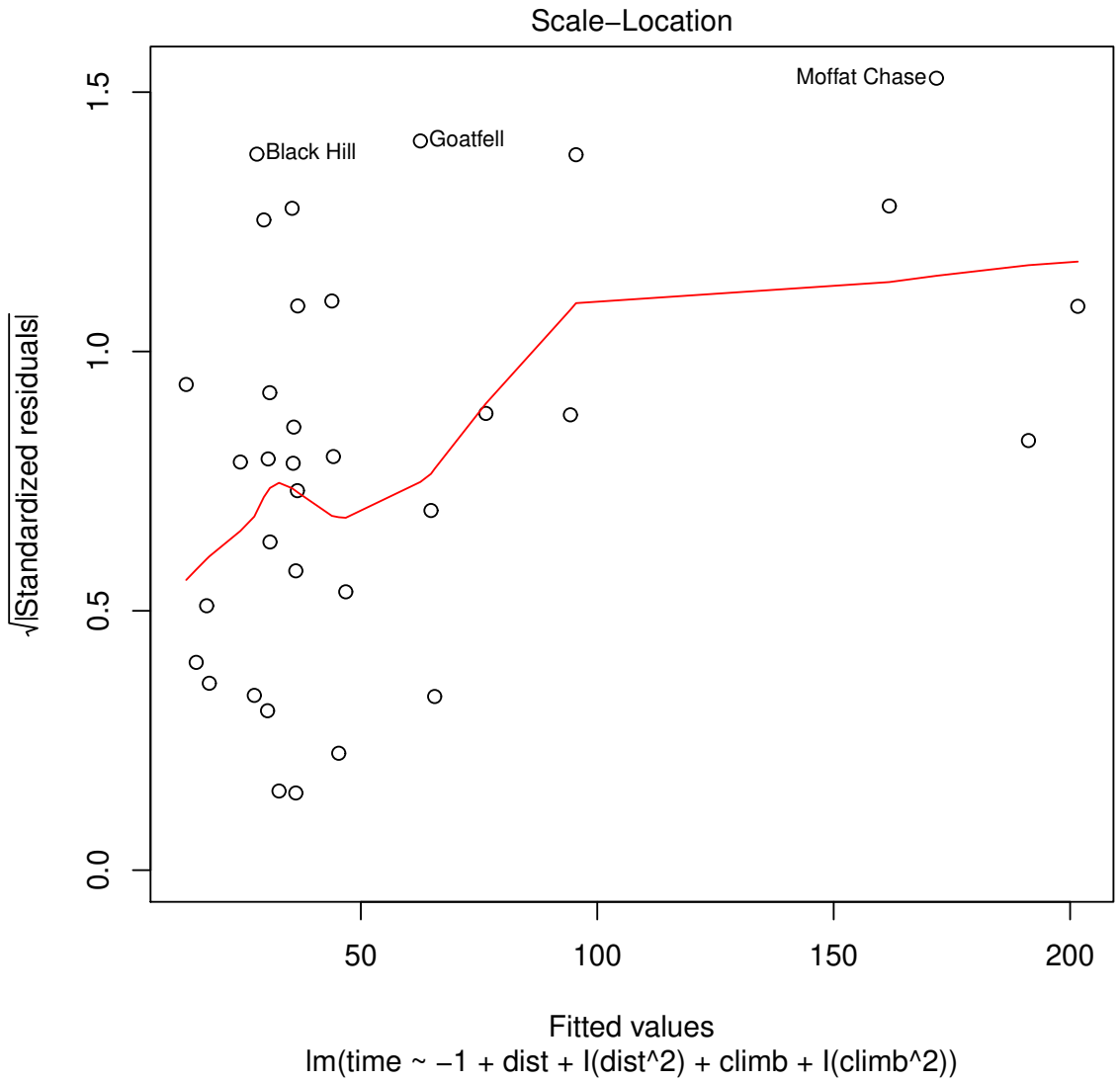
$$\begin{aligned}\widehat{\text{time}} &= 5.62 \times \text{distance} + 0.0323 \times (\text{distance})^2 \\ &\quad + 0.000262 \times \text{climb} + 0.00000180 \times (\text{climb})^2\end{aligned}$$

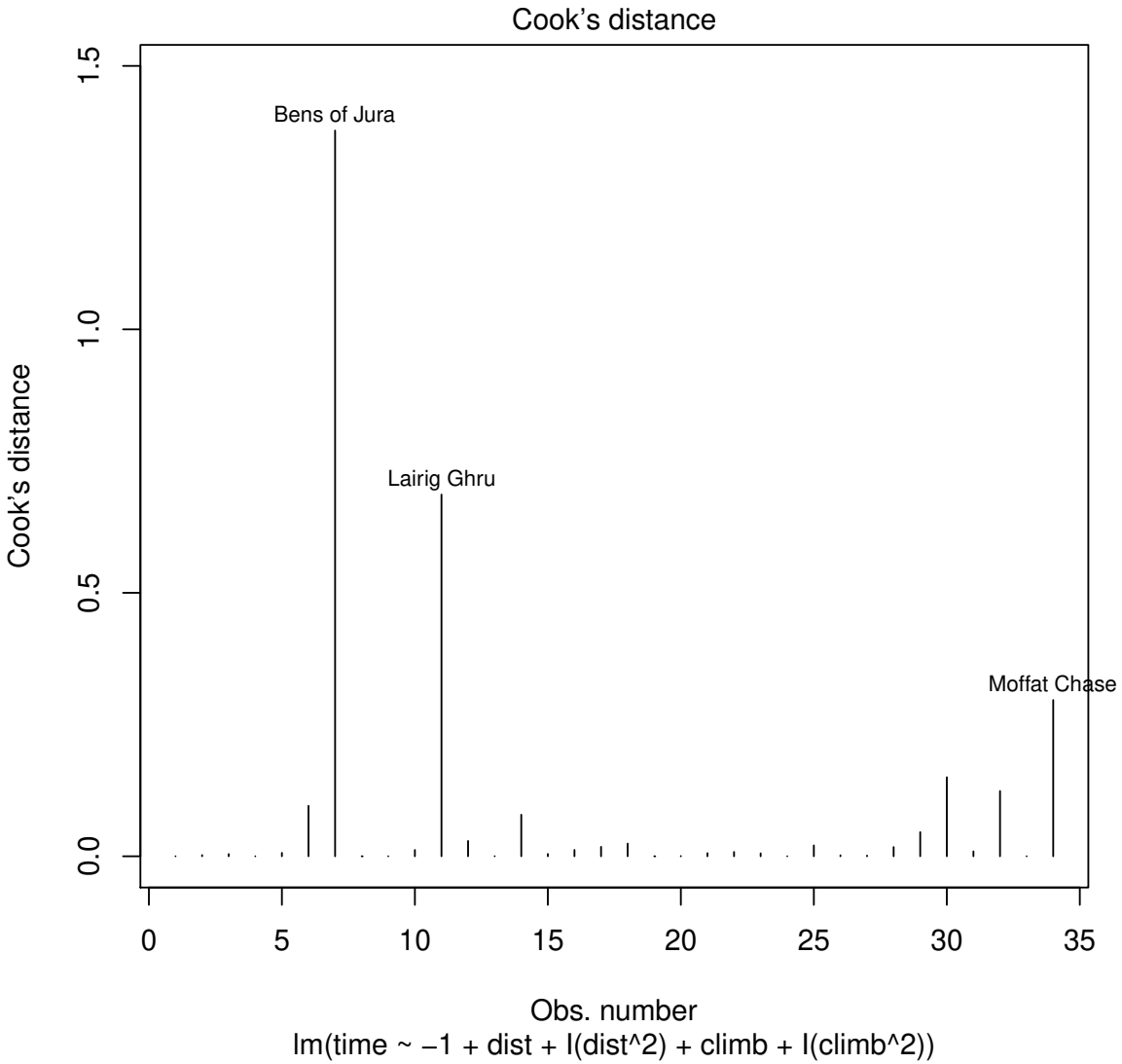
Analysis shows:

- Evidence of nonlinearity in the dependence on *climb* — see coefficient of $(\text{climb})^2$, and associated p-value
- Quite possibly in the dependence on distance (given also physical considerations)


Residuals vs Fitted
Goatfell
Black Hill
Moffat Chase
Residuals
10
5
0
−5
−10
50
100
150
200
Fitted values
lm(time ~ −1 + dist + I(dist^2) + climb + I(climb^2))

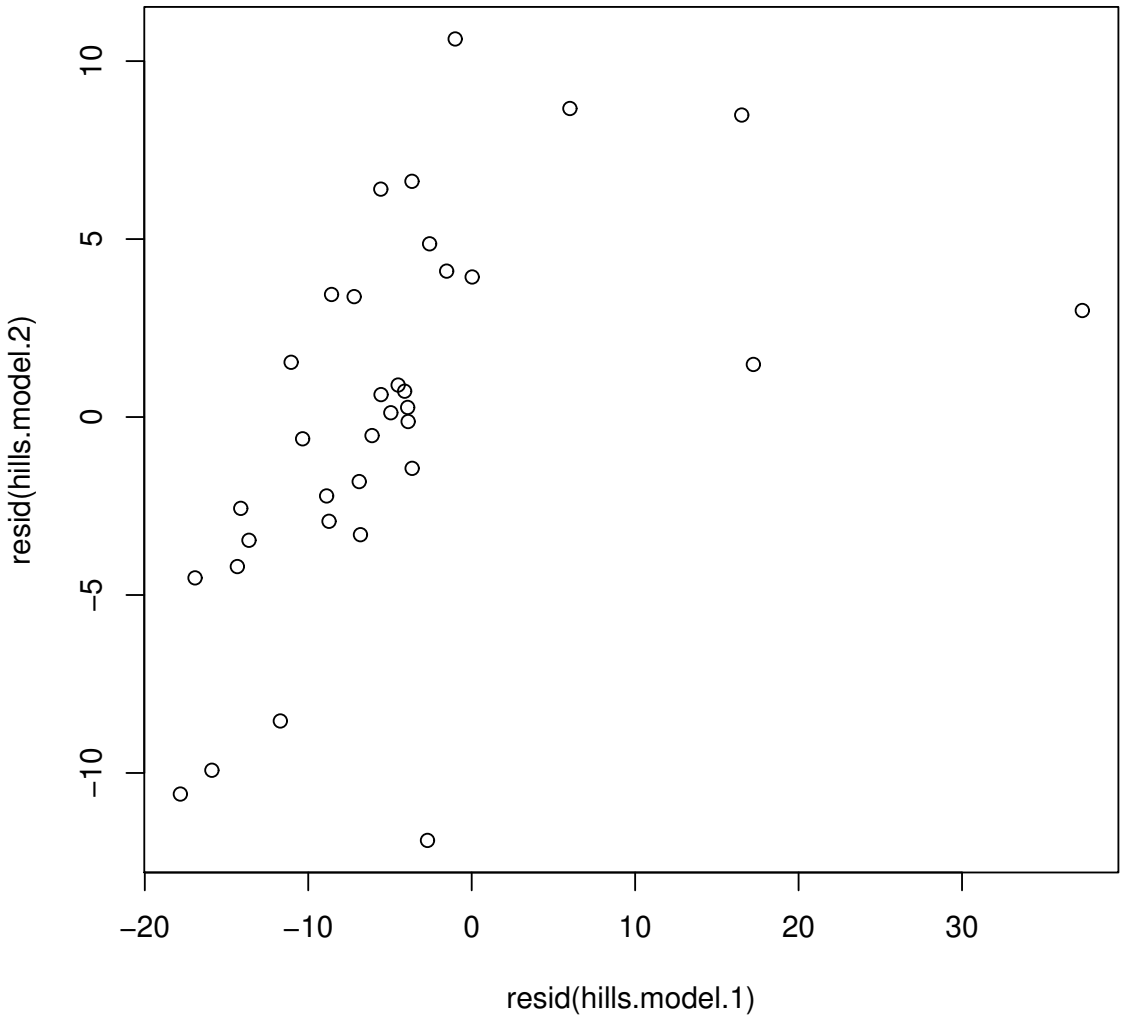

Normal Q–Q
Standardized residuals
Theoretical Quantiles
lm(time ~ –1 + dist + I(dist^2) + climb + I(climb^2))
Goatfell
Black Hill
Moffat Chase
2
1
0
–1
–2
–2
–1
0
1
2


Scale−Location
Moffat Chase
Black Hill
Goatfell
√|Standardized residuals|
0.0
0.5
1.0
1.5
50
100
150
200
Fitted values
lm(time ~ −1 + dist + I(dist^2) + climb + I(climb^2))


Cook's distance
Bens of Jura
Lairig Ghru
Moffat Chase
Cook's distance
0.0
0.5
1.0
1.5
0
5
10
15
20
25
30
35
Obs. number
lm(time ~ −1 + dist + I(dist^2) + climb + I(climb^2))

Can also plot residuals of Model 2 against those of Model 1.

```
> plot(resid(hills.model.2) ~ resid(hills.model.1))
```



This suggests that Model 2 is an improvement – it reduces the size of the largest residuals.

The observations corresponding to Bens of Jura and Lairig Ghru remain moderately influential.